

Fraol Batole

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PhD Researcher — Software Engineering for AI | Coding Agents | Program Analysis

EDUCATION

Tulane University — Ph.D., Computer Science

Aug 2024 – Present

New Orleans, Louisiana | GPA: 4.0/4.0 | Advisor: Prof. Hridesh Rajan

Iowa State University — M.Sc., Computer Science

May 2024

Ames, Iowa | GPA: 3.62/4.0

Addis Ababa Science and Technology University — B.Sc., Software Engineering

June 2018

Addis Ababa, Ethiopia | GPA: 3.75/4.0 | Graduated Magna Cum Laude | Founded Innovation Club

RESEARCH EXPERIENCE

Laboratory for Software Engineering, Tulane University

Aug 2024 – Present

Graduate Research Assistant

New Orleans, Louisiana

- **LLM Agent Reliability (RefineAct):** Designing a runtime verification framework that automatically derives task-specific formal specifications and enforces them during LLM agent execution, catching errors that static analysis and prompting alone cannot (under submission, ASE '26).
- **AST-Aware Pretraining (CAPS):** Proposed an AST-aware pretraining strategy using type annotations and syntactic structure to substitute for raw parameter scale in code models, improving code generation without scaling compute (under submission, TSE).
- **Agent-Oriented Fault Localization (LocalizeAgent):** Built a multi-agent framework integrating static analysis to produce abstraction-aware, LLM-friendly code summaries for automated design issue localization — accepted at ICSE '25.
- **LLM Agent Bug Study:** Led the first large-scale empirical study of LLM-based agent bugs, analyzing 1,016 bug reports and identifying tool integration (45%) and model integration (16%) as primary failure modes (under submission).
- **Move Method Refactoring (MM-Assist) — JetBrains Collaboration:** Developed and released MM-Assist, an IntelliJ IDEA plugin for automated Move Method refactoring using LLMs, RAG, and hallucination filtering — achieving 73% accuracy and 4× improvement over prior tools (ICSME '25).
- **Typestate-Based Fault Localization:** Developed static analysis techniques to detect API usage violations in deep learning programs using typestate analysis, in collaboration with IBM (TSE '26).

Polytechnique Montréal — Research Intern

May 2025 – Aug 2025

Supervisor: Prof. Foutse Khomh

Montreal, Canada

- Investigated safety and robustness of LLM-based AI agents, focusing on novel detection and mitigation strategies for agent misbehavior; work directly informs RefineAct (under submission, ASE '26).

TECHNICAL SKILLS

Programming Languages: Python, Java

ML / LLM Frameworks: PyTorch, TensorFlow/Keras, Hugging Face Transformers, fine-tuning, RAG pipelines

Agent & Tooling: LangChain, LangGraph, CrewAI, multi-agent frameworks, MCPs

Program Analysis / SE: Static analysis, AST manipulation, typestate analysis, runtime verification, JetBrains Platform SDK (IntelliJ plugin development)

Cloud / Infrastructure: Google Cloud Platform (1.5M NSF credits + \$12K GCP research grant), distributed training, HPC clusters

SELECTED PUBLICATIONS (FIRST-AUTHOR)

Accepted

1. **Fraol Batole**, Ruchira Manke, Tien N. Nguyen, Robert Dyer, and Hridesh Rajan. "Typestate-Based Fault Localization of API Usage Violations in Deep Learning Programs." *IEEE Transactions on Software Engineering (TSE '26)*.
2. **Fraol Batole**, David O'Brien, Tien N. Nguyen, Robert Dyer, and Hridesh Rajan. "An LLM-Based Agent-Oriented Approach for Automated Code Design Issue Localization." *ICSE '25*.

Under Submission

3. **Fraol Batole**, Foutse Khomh, and Hridesh Rajan. "Automatic Runtime Verification of LLM Agent Actions"
4. **Fraol Batole**, Smit Patel, Aashish Yadavally, Tien N. Nguyen, and Hridesh Rajan. "CAPS: Code Abstraction Pre-training Strategy for Large Language Models."
5. **Fraol Batole**, Ruchira Manke, David O'Brien, Steven Vo, Tien N. Nguyen, and Hridesh Rajan. "A Comprehensive Study on LLM-Based Agent Bug Characteristics."

Full Publications & Co-authored Work

Accepted

6. Abhiram Bellur, Mohammed Raihan Ullah, **Fraol Batole** et al. "Multi-Agent Coordinated Rename Refactoring." *FSE'26 (Tool Demo Track)*
7. Abhiram Bellur, **Fraol Batole** et al. "LLM, IDE and Semantic Embedding to Assist Move Method Refactoring." *ICSME '25*.
8. Sayem Imtiaz, Astha Singh, **Fraol Batole**, and Hridesh Rajan. "IRepair: An Intent-Aware Approach to Repair Data-Driven Errors in LLMs." *FSE '25*.
9. Sayem Imtiaz, **Fraol Batole** et al. "Decomposing RNN to Enable Reuse." *ICSE '23*.
10. Abhiram Bellur, **Fraol Batole**. "IDE Native, Foundation Model-Based Agents for Software Refactoring." *IDE Workshop @ ICSE '25*.

Under Submission

11. Samuel W. Flint, Robert Dyer, **Fraol Batole**, and David O'Brien. "Do LLMs Generate Code with Type Annotations?"
12. David O'Brien, **Fraol Batole** et al. "Issue-Based Programming Language Evolution."

ACADEMIC SERVICE & AWARDS

Distinguished Reviewer Award (Junior PC) — MSR '26

April 2026

Reviewer / Program Committee:

- Junior Program Committee (Technical Track) — Mining Software Repositories (MSR '26)
- Reviewer — Transactions on Software Engineering and Methodology (TOSEM '26)
- Reviewer — Automated Software Engineering (ASE) Journal '25
- Reviewer — Deep Learning for Code Workshop @ NeurIPS '25

ADDITIONAL EXPERIENCE

Iowa State University — Graduate Teaching Assistant

Jan 2021 – Aug 2024

CS Department

Ames, Iowa

- Taught programming principles, debugging, and test-case development to 80+ students; adapted pedagogy for diverse learning styles.

MED Innovation — Software Engineer

Aug 2019 – Nov 2019

Addis Ababa, Ethiopia

- Integrated an existing CRM system for a cross-functional team using Spring Boot and Vaadin.

HONORS & GRANTS

- **NSF-Funded Advanced Computing Credits:** 1.5 million credits awarded (Feb 2024, Jun 2025).
- **Google Cloud Research Credits:** \$12,000 awarded (Jun 2023).
- **Google Cloud Research Innovator '23:** Competitively selected for Google's research program (Mar 2023).
- **Magna Cum Laude, B.Sc. Software Engineering** — Addis Ababa Science and Technology University.